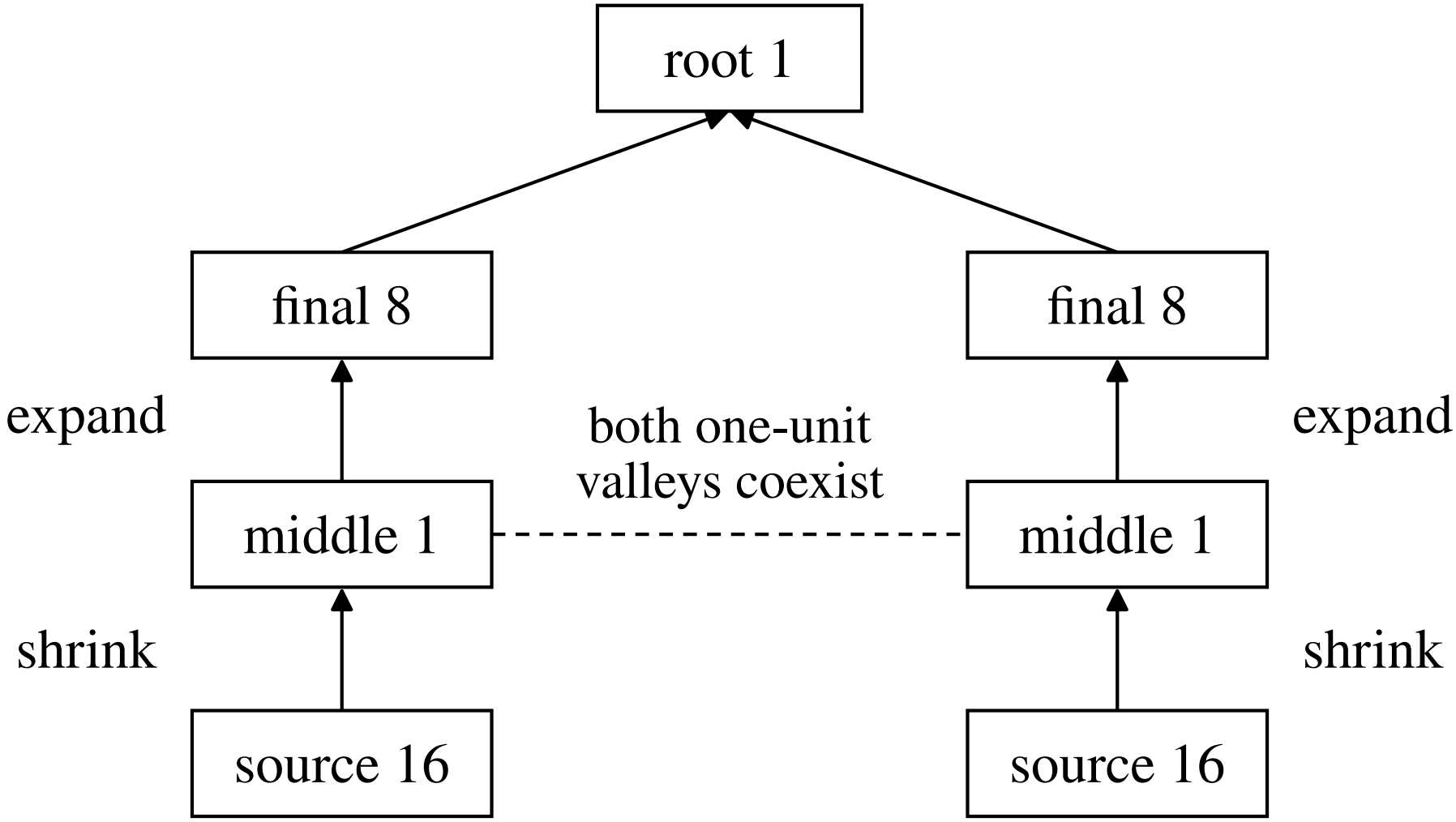

$B = 18$; optimum I/O $= 16 + 16 + 1 = 33$